

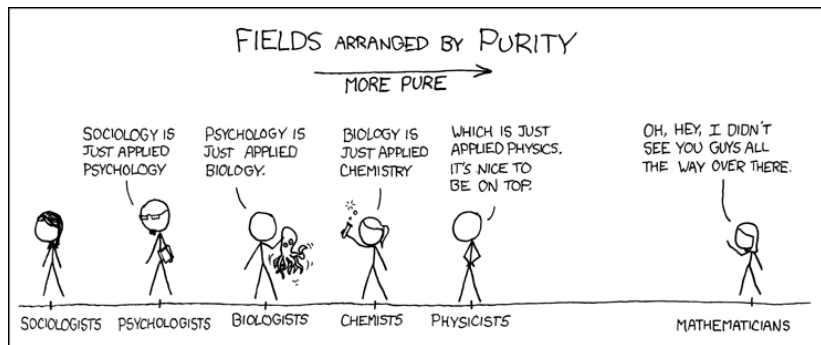
Algorithmic and artificial intelligence approaches to the latest bioinformatics challenges

Murray Patterson

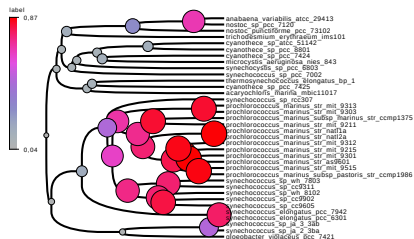
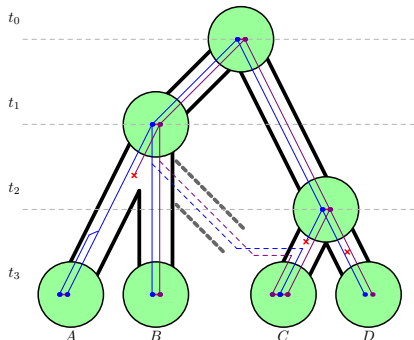
Department of Computer Science, Georgia State University

Bio for Breakfast — March 2026





Co-evolution of genes

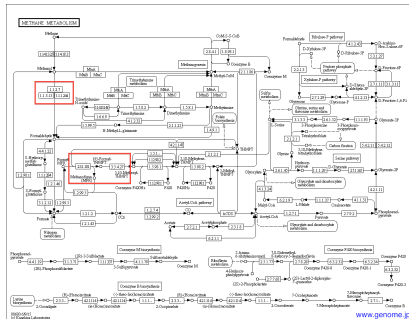
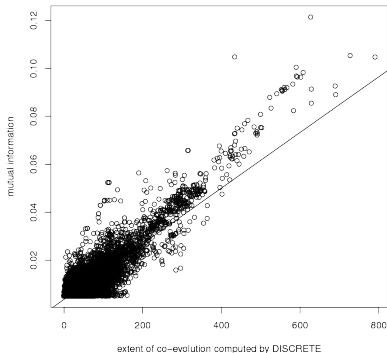


with Eric Tannier *et al.*, at the LBBE

- ▶ DeCoLT (Patterson *et al.*, BMC Bioinformatics, 2013)
pbil.univ-lyon1.fr/software/DeCoLT/
- ▶ DeCoSTAR (Duchemin *et al.*, Genome Biology and Evolution, 2017)
pbil.univ-lyon1.fr/software/DeCoSTAR/

Correlated evolution of proteins

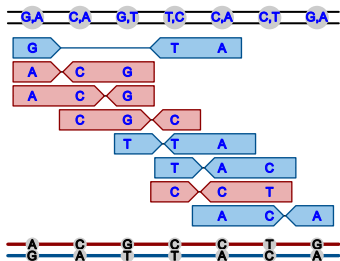
Comparison of mutual information with DISCRETE



with Daniel Kahn *et al.*, at the LBBE

- ▶ (Patterson *et al.*, Society for Molecular Biology and Evolution, 2015)
- ▶ (Patterson *et al.*, bioRxiv, 2016)

Haplotype assembly



- ▶ important for population genetics (Browning and Browning, 2011) and clinical decisions (Glusman et al., 2014)
- ▶ hard optimization problem (Cilibrasi et al., 2005)

with Alexander Schönhuth et al., at the CWI

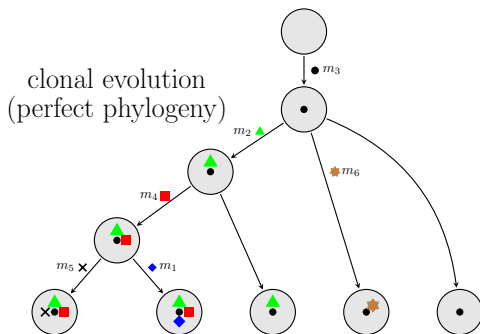
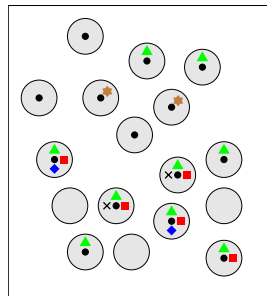
- ▶ WhatsHap (Patterson and Marshall et al., Recomb/JCB, 2015)
- ▶ whatshap (tool) (Martin and Patterson et al., bioRxiv, 2016)
whatshap.readthedocs.io

with Gianluca Della Vedova, Paola Bonizzoni et al., at the Algotlab

- ▶ HapCHAT (Beretta and Patterson et al., BMC Bioinformatics, 2018)
hapchat.algotlab.eu
- ▶ HapQual <https://github.com/AlgoLab/HapQual> —
collaboration with Justin Zook et al., at the NIST

Inferring tumor phylogeny

observation: cells



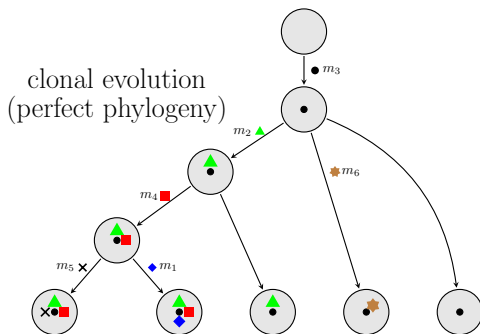
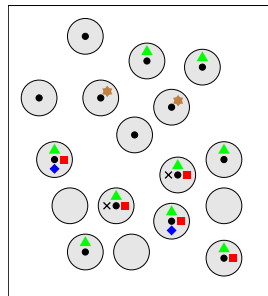
Drawbacks of single cell

- ▶ still suffers from moderate dropout and false negative rates
— due to low coverage
- ▶ more expensive than high throughput bulk sequencing

m_1	m_2	m_3	m_4	m_5	m_6
0	0	0	0	0	0
0	0	1	0	0	0
0	1	1	0	0	0
0	1	1	1	0	0
0	1	1	1	1	0
1	1	1	1	0	0
0	0	1	0	0	1

Inferring tumor phylogeny

observation: cells

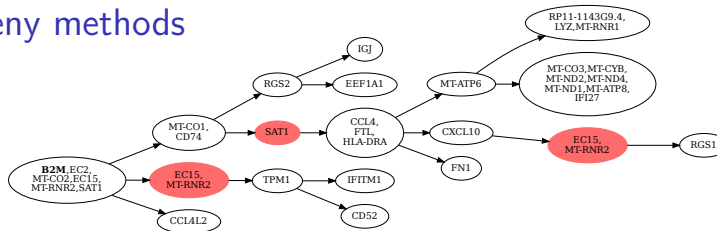


Drawbacks of single cell

- ▶ still suffers from moderate dropout and false negative rates
— due to low coverage
- ▶ more expensive than high throughput bulk sequencing

m_1	m_2	m_3	m_4	m_5	m_6
2	0	0	0	0	0
0	0	2	0	0	2
0	0	1	2	0	0
2	1	2	1	2	0
0	0	1	0	1	2
1	1	0	1	2	0
0	2	1	1	0	2

Tumor phylogeny methods



gpps: general parsimony phylogeny from single cell

(Ciccolella *et al.*, ICCABS 2018) github.com/AlgoLab/gppf — a method for inferring a Dollo(k) phylogeny using *integer linear programming* (ILP)

sasc: simulated annealing single-cell

(Ciccolella *et al.*, Recomb-CCB 2018; bioRxiv 268243)

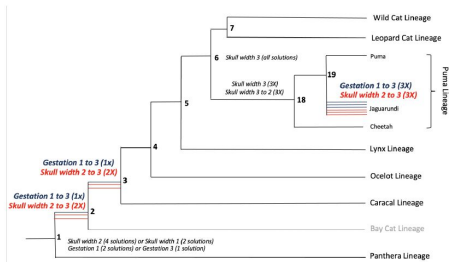
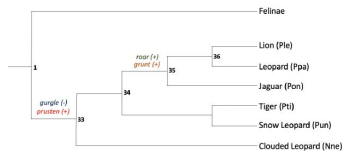
omictools.com/sasc-tool — collaboration with Dana Silverbush and Mario Suva at the Broad Institute

Celluloid

(Ciccolella, Patterson *et al.*, ACM-BCB 2019)

github.com/AlgoLab/celluloid — a technique for clustering (columns) of an SCS matrix

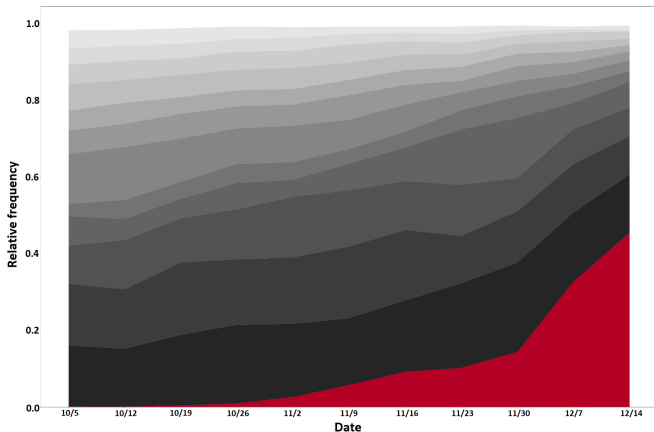
Correlated evolution of felid vocalizations



with S. Ashley Byun and Danushka Bandara *et al.*, at Fairfield University

- ▶ **parcours** (Smith *et al.*, ICCABS 2021)
<https://github.com/murraypatterson/parcours>
- ▶ (Khandai *et al.*, JCB, 2022) — computing significant correlation
- ▶ (Bandara *et al.*, Acoustics Australia, 2023) — classifying species and call-type

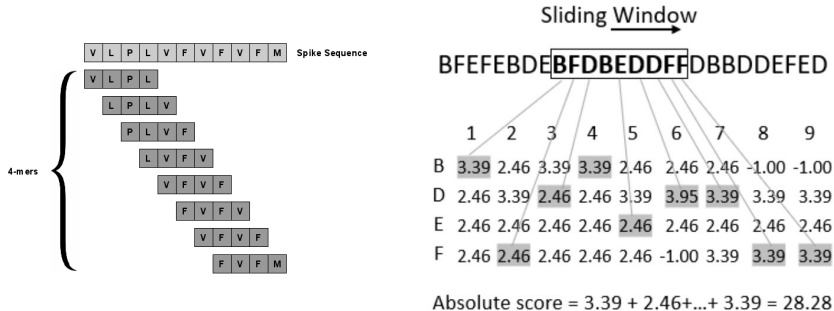
Clustering SARS-CoV-2 subtypes



with Alex Zelikovsky and Pavel Skums *et al.*, at GSU and UConn

► (Melnyk *et al.*, ICCABS 2020 & JCB, 2021)

Numerically embedding SARS-CoV-2 sequences

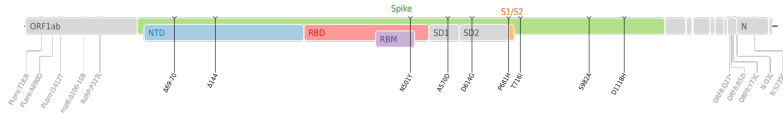
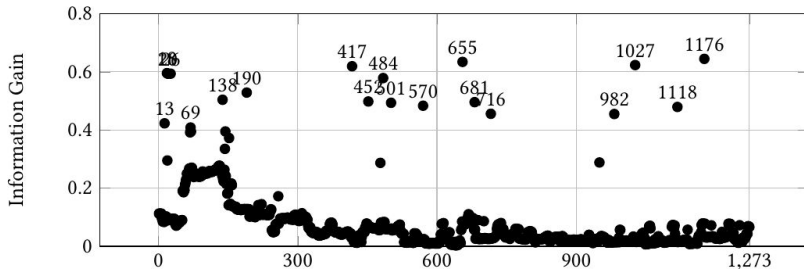


with my research group at GSU

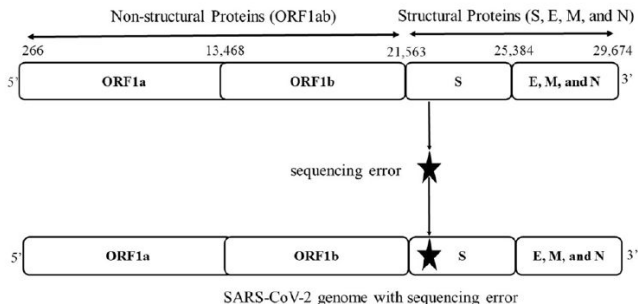
- ▶ (Ali & Patterson, IEEE BigData 2021); (Ali *et al.*, ICBDR 2021); (Ali *et al.*, ISBRA 2021); (Tayebi *et al.*, MDPI Algorithms, 2021); (Ali *et al.*, MDPI Biology, 2022); (Ali *et al.*, ICONIP 2022); (Ali *et al.*, CHIL 2023); (Ali *et al.*, MBEC, 2023); (Ali *et al.*, ISBRA 2023); (Ali *et al.*, PAKDD 2023); (Ali & Murad *et al.*, IJCNN 2023); ...

Interpreting the numerical embeddings

in (Ali & Patterson, IEEE BigData 2021):



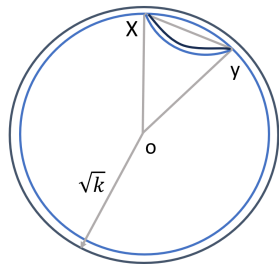
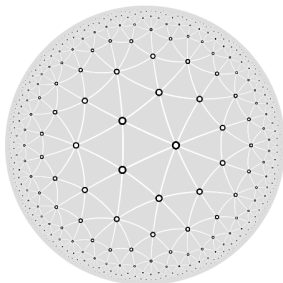
Benchmarking robustness of the numerical embeddings



with Sarwan Ali, Bikram Sahoo, and Pinyu Chen *et al.*, at GSU and IBM

- ▶ (Ali *et al.*, Scientific Reports, 2023)
- ▶ (Ali *et al.*, ISBRA 2023)
- ▶ (Sahoo *et al.*, Biomolecules, 2023)

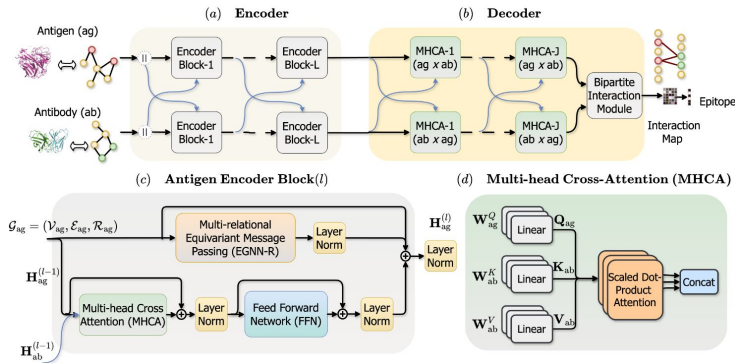
Hyperbolic embeddings



with my research group at GSU

- ▶ (Ali *et al.*, Knowledge-Based Systems, 2024)
- ▶ (Ali *et al.*, ISBRA 2024)
- ▶ (Ali *et al.*, SIMBig 2024)
- ▶ (Chourasia, Ali & Patterson, Submitted to IJCNN 2026)

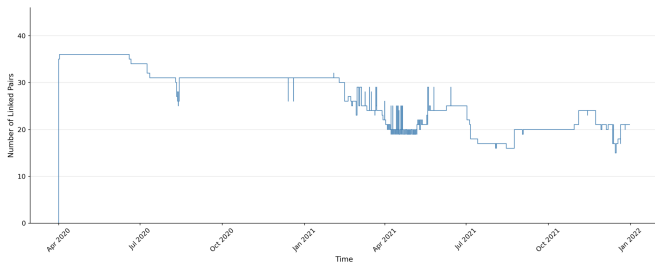
Antibody design



with my research group at GSU, and Imdad Khan *et al.*, at LUMS

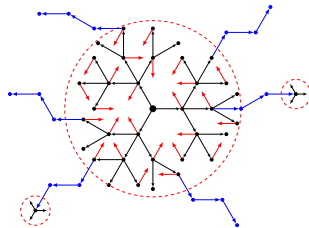
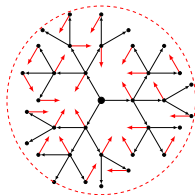
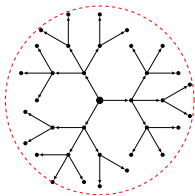
- (Sardar *et al.*, ISBRA 2023) — nanobody-antigen binding prediction
- (Ahmed *et al.*, ICCABS 2025) — antibody-aware epitope prediction
- (Ahmed *et al.*, GEM ICLR 2026) — Chimera-Bench system
- (Ahmed *et al.*, ICCABS 2026) — benchmarking antibody design

Patterns of evolution from sequencing big data

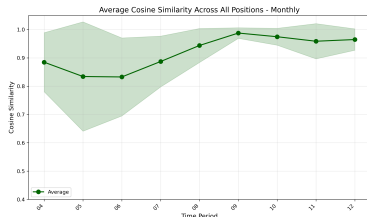
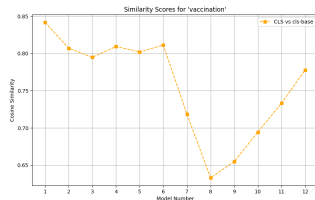


with my research group, Alex Zelikovsky (GSU), Pavel Skums (UConn)

► (Adeniyi *et al.*, JCB, 2026)

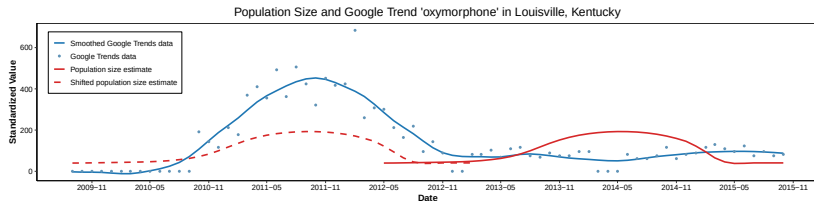


Semantic shifts



with Mario Kubek, Alexander Kirpich, Gerardo Chowell, and Alex Zelikovsky at GSU, and Pavel Skums at UConn

► (Guntupalli *et al.*, ICNLP 2026)



Thank you!

